Modular Inverses and Congruences

COMP2711 Discrete Mathematical Tools

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Contents

1. Modular inverses and equations
2. Combining congruences
3. Modular powers: Fermat and Euler
4. Optional: Non-coprime equations
5. Review exercises

Review: Congruence and arithmetic

All numbers are integers; throughout this lecture, moduli are greater than 1.

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If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a \pm c \equiv b \pm d \pmod{m}, \qquad ac \equiv bd \pmod{m}.$$

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Division is not always valid:

$$2 \times 1 \equiv 2 \times 4 \pmod{6}, \qquad 1 \not\equiv 4 \pmod{6}.$$

Review: GCD and Euclid

$\gcd(a, b)$ is the greatest positive common divisor of a, b (not both zero).
If $\gcd(a, b) = 1$, we say a, b are **coprime**.

Euclidean Algorithm for GCD:

$$\gcd(a, b) = \gcd(a \bmod b, b) \qquad (a \geq 0,\ b > 0).$$

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Can we solve this equation?

In elementary algebra, over the real numbers:

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But $\div$ does not preserve congruence in general.

What should "divide by 17" mean modulo 43?

Modular inverse

MUST UNDERSTAND Modular inverse

An **inverse** of a modulo m is an integer u such that

$$au \equiv 1 \pmod{m}.$$

Notation: $u \equiv a^{-1} \pmod{m}$.

This plays the role of $1/a$ in modular arithmetic, when it exists.

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Example: $3 \times 5 = 15 \equiv 1 \pmod{7}$, so $5 \equiv 3^{-1} \pmod{7}$.

How do we find a modular inverse?

Given a and m, we want an integer u such that $au \equiv 1 \pmod{m}$.

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If $\gcd(a, m) = 1$*, there exist integers* u, k *such that*

$$au + km = 1.$$

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Run Euclid first; then work backward to recover the coefficients.

Extended Euclid: 1. Run Euclid

a	b
43	17

Recall

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

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a	b
43	17
17	$9 = 43 \bmod 17$

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1	$0 = 8 \bmod 1$

Recall

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Extended Euclid: 2. The key backward step

Suppose we know a Bézout solution pair—integers s, t—for the smaller pair $(b, a \bmod b)$, such that

$$sb + t(a \bmod b) = 1.$$

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$$ta + (s - tq)b = 1.$$

Extended Euclid: 2. The key backward step

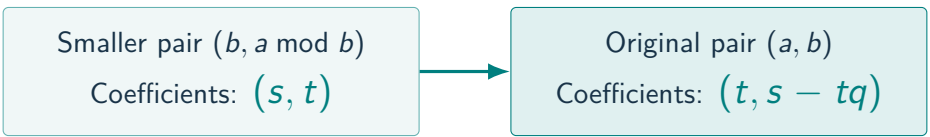
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Write $a = qb + r$, where $r = (a \bmod b) = a - qb$.

$$sb + t(a - qb) = 1.$$

$$ta + (s - tq)b = 1.$$



Extended Euclid: 3. Work backward through the table

In each row, find s, t such that $sa + tb = 1$.

a	b	$q = \lfloor a/b \rfloor$	s	t
43	17	2		
17	9	1		
9	8	1		
8	1	8		
1	0	—		

Original pair (a, b)

Coefficients: $(t, s - tq)$



Smaller pair $(b, a \bmod b)$

Coefficients: (s, t)

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1	**0**	—	1	0

Start at $(1, 0)$: choose $(s, t) = (1, 0)$.

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$$1 \times 1 + 0 \times 0 = 1.$$

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1	0	—	1	**0**

Using $q = 8$: $(s, t) = (0, 1 - 0 \times 8) = (0, 1)$.

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43	17	2		
17	9	1		
9	**8**	1	**1**	-1
8	1	8	0	**1**
1	0	—	1	0

Using $q = 1$: $(s, t) = (1, 0 - 1 \times 1) = (1, -1)$.

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Using $q = 1$: $(s, t) = (1, 0 - 1 \times 1) = (1, -1)$.

$$1 \times 9 + (-1) \times 8 = 1.$$

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Using $q = 1$: $(s, t) = (-1, 1 - (-1) \times 1) = (-1, 2)$.

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Using $q = 1$: $(s, t) = (-1, 1 - (-1) \times 1) = (-1, 2)$.

$$(-1) \times 17 + 2 \times 9 = 1.$$

Extended Euclid: 3. Work backward through the table

In each row, find s, t such that $sa + tb = 1$.

a	b	$q = \lfloor a/b \rfloor$	s	t
43	**17**	2	**2**	-5
17	9	1	-1	**2**
9	8	1	1	-1
8	1	8	0	1
1	0	—	1	0

Using $q = 2$: $(s, t) = (2, -1 - 2 \times 2) = (2, -5)$.

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Using $q = 2$: $(s, t) = (2, -1 - 2 \times 2) = (2, -5)$.

$$2 \times 43 + (-5) \times 17 = 1.$$

Extended Euclid: 4. Read the answer

The top row gives

$$2 \times 43 - 5 \times 17 = 1.$$

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Return to our original goal: find u such that $17u \equiv 1 \pmod{43}$. We now know

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Return to our original goal: find u such that $17u \equiv 1 \pmod{43}$. We now know

$$(-5) \times 17 = 1 - 2 \times 43.$$

$$(-5) \times 17 \equiv 1 \pmod{43}.$$

Thus our inverse is

$$\boxed{-5 \equiv 38 \equiv 17^{-1} \pmod{43}.}$$

Bézout's identity

Theorem (Bézout)

For integers a, m not both zero, there exist integers u, k such that

$$au + km = \gcd(a, m).$$

*The **Extended Euclidean Algorithm** finds such a pair (u, k).*

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The ***Extended Euclidean Algorithm*** *finds such a pair (u, k).*

The proof is nearly identical to the extended Euclidean argument: work backward from $(g, 0)$, where $g = \gcd(a, m)$.

The full proof is in the course note.

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Corollary (Modular inverses and coprimality)

For an integer a and $m > 1$,

$$\begin{aligned} & a \text{ has an inverse modulo } m \\ \iff\ & \text{there exist integers } u, k \text{ such that } au + km = 1 \\ \iff\ & \gcd(a, m) = 1. \end{aligned}$$

Solving the opening equation

$$17x \equiv 5 \pmod{43}, \qquad 17^{-1} \equiv 38 \pmod{43}.$$

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Multiply both sides by 38:

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$$\boxed{x = 18 + 43k, \qquad k \in \mathbb{Z}.}$$

Check: $17 \times 18 = 306 = 7 \times 43 + 5$.

Cancellation with an inverse

If $\gcd(c, m) = 1$, then

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Why the inverse is unique modulo m

If $au \equiv 1$ and $av \equiv 1 \pmod{m}$, then

$$au \equiv av \pmod{m} \implies u \equiv v \pmod{m}.$$

Non-coprime equations: The idea

If $g = \gcd(a, m) > 1$, try reducing to the coprime case.

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If $g = \gcd(a, m) > 1$, try reducing to the coprime case.

For example, divide the coefficient, right-hand side, and modulus by 2:

$$6x \equiv 8 \pmod{14} \iff 3x \equiv 4 \pmod{7}.$$

Now $\gcd(3, 7) = 1$, so we can use a modular inverse.

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The full procedure and solution set are in the optional material at the back.

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Sunzi's remainder problem

《孫子算經》卷下，第26題

Traditional Chinese	Mathematical formulation
今有物不知其數，	Let x be the number of objects.
三三數之賸二，	$x \equiv 2 \pmod 3$.
五五數之賸三，	$x \equiv 3 \pmod 5$.
七七數之賸二，	$x \equiv 2 \pmod 7$.
問物幾何？	Find x.

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答曰：「二十三。」 Answer: 23.

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Chinese remainder theorem

MUST UNDERSTAND Chinese remainder theorem

Let $m_1, \ldots, m_n > 1$ be pairwise coprime: $\gcd(m_i, m_j) = 1$ for $i \neq j$. For any integers $r_1, \ldots, r_n$, the system

$$\begin{aligned} x &\equiv r_1 \pmod{m_1}, \\ &\vdots \\ x &\equiv r_n \pmod{m_n} \end{aligned}$$

has exactly one solution modulo $M = m_1 \cdots m_n$.

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We start with the case of two congruences.

Chinese remainder theorem: Two congruences

Theorem

Let $m_1, m_2 > 1$ *with* $\gcd(m_1, m_2) = 1$. *For any integers* r_1, r_2, *the system*

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For more congruences: Combine the 1st and 2nd, then combine with the 3rd, the 4th, ...

CRT construction: 1. Use Bézout's identity

Bézout gives integers u_1, u_2 with

$$u_1 m_1 + u_2 m_2 = 1.$$

$$\gcd(m_1, m_2) = 1$$
$$x \equiv r_1 \pmod{m_1}$$
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Multiply by r_2:

$$r_2 u_1 m_1 + r_2 u_2 m_2 = r_2.$$

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Modulo m_2, the second term vanishes:

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Raw term	Modulo m_1	Modulo m_2
$r_2 u_1 m_1$	0	r_2

CRT construction: 2. Match the remainders

Similarly, multiply Bézout's identity by r_1:

$$r_1 u_1 m_1 + r_1 u_2 m_2 = r_1.$$

$$\gcd(m_1, m_2) = 1$$
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Raw term	Modulo m_1	Modulo m_2
$r_2 u_1 m_1$	0	r_2
$r_1 u_2 m_2$	r_1	0
$x \equiv r_2 u_1 m_1 + r_1 u_2 m_2 \pmod{m_1 m_2}$	r_1	r_2

CRT construction: 3. Work through the example

$$x \equiv 2 \pmod 3,$$
$$x \equiv 3 \pmod 5.$$

$\gcd(m_1, m_2) = 1$
$x \equiv r_1 \pmod{m_1}$
$x \equiv r_2 \pmod{m_2}$

CRT construction: 3. Work through the example

$$x \equiv 2 \pmod 3,$$
$$x \equiv 3 \pmod 5.$$

Bézout: $2 \times 3 + (-1) \times 5 = 1$, so $u_1 = 2$, $u_2 = -1$.

$\gcd(m_1, m_2) = 1$
$x \equiv r_1 \pmod{m_1}$
$x \equiv r_2 \pmod{m_2}$

Raw term	Modulo 3	Modulo 5
$r_2 u_1 m_1 = 3 \times 2 \times 3 = 18$	0	3
$r_1 u_2 m_2 = 2 \times (-1) \times 5 = -10$	2	0

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Raw term	Modulo 3	Modulo 5
$r_2 u_1 m_1 = 3 \times 2 \times 3 = 18$	0	3
$r_1 u_2 m_2 = 2 \times (-1) \times 5 = -10$	2	0

$$x \equiv 18 + (-10) = 8 \pmod{15}.$$

CRT construction: 3. Work through the example

$$x \equiv 2 \pmod 3,$$
$$x \equiv 3 \pmod 5.$$

Bézout: $2 \times 3 + (-1) \times 5 = 1$, so $u_1 = 2$, $u_2 = -1$.

$\gcd(m_1, m_2) = 1$
$x \equiv r_1 \pmod{m_1}$
$x \equiv r_2 \pmod{m_2}$

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$$x \equiv 18 + (-10) = 8 \pmod{15}.$$

All integer solutions: $x = 8 + 15k$, $k \in \mathbb{Z}$.

CRT: Why the solution is unique

Assume x and y are any two solutions:

$$x \equiv r_1 \pmod{m_1} \quad y \equiv r_1 \pmod{m_1}$$
$$x \equiv r_2 \pmod{m_2} \quad y \equiv r_2 \pmod{m_2}$$

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$$m_1 \mid (x - y), \qquad m_2 \mid (x - y).$$

That is, m_1 and m_2 are both divisors of $(x - y)$.

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Lemma

If a and b are coprime, i.e., $\gcd(a, b) = 1$, and both divide x, then $ab \mid x$.

Proof: exercise; see the course note.

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Lemma

If a and b are coprime, i.e., $\gcd(a, b) = 1$, and both divide x, then $ab \mid x$.

Proof: exercise; see the course note.

Thus $m_1 m_2 \mid (x - y)$, so $x - y \equiv 0 \pmod{m_1 m_2}$.

Optional: The general CRT solution

For pairwise-coprime $m_1, \ldots, m_n > 1$, solve $x \equiv r_i \pmod{m_i}$.

$$M = \prod_{i=1}^{n} m_i, \qquad M_i = \frac{M}{m_i}.$$

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General solution

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Modulo m_j, only the jth term survives, giving remainder r_j.

Contents

1. Modular inverses and equations

2. Combining congruences

3. **Modular powers: Fermat and Euler**

4. Optional: Non-coprime equations

5. Review exercises

Review: Computing modular powers

Compute

$$2^{13} \equiv ? \pmod 5.$$

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Cycles

$$2 \to 4 \to 3 \to 1 \to 2 \pmod 5$$

$$2^{13} = (2^4)^3\, 2 \equiv 2 \pmod 5.$$

Repeated squaring

$$13 = 8 + 4 + 1$$

$$2^{13} = 2^8\, 2^4\, 2 \equiv 1 \times 1 \times 2 \pmod 5.$$

Review: Computing modular powers

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Repeated squaring

$$13 = 8 + 4 + 1$$

$$2^{13} = 2^8\, 2^4\, 2 \equiv 1 \times 1 \times 2 \pmod 5.$$

Can a theorem give a valid exponent reduction without first building the cycle?

Fermat's little theorem

MUST UNDERSTAND Modular powers

If p is **prime** and $1 \leq a \leq p-1$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Fermat's little theorem

MUST UNDERSTAND Modular powers

If p is **prime** and $1 \leq a \leq p-1$, then

$$a^{p-1} \equiv 1 \pmod{p}.$$

Example:

$$3^6 \equiv 1 \pmod{7}.$$

Equivalent all-integer form:

$$a^p \equiv a \pmod{p} \qquad \text{for every integer } a.$$

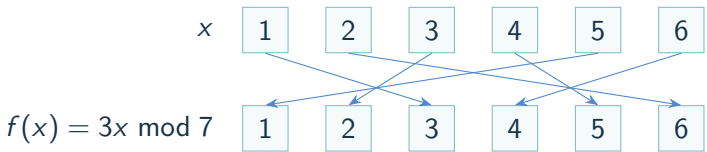
Fermat's Little Theorem: proof

p prime, $1 \leq a \leq p-1$
$a^{p-1} \equiv 1 \pmod{p}$

Consider $S = \{1, 2, \ldots, p-1\}$ and

$$f : S \to S, \qquad f(x) = ax \bmod p.$$

Example: $p = 7$, $a = 3$.



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f is a bijection (a one-to-one correspondence).

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Hint: Since $\gcd(a, p) = 1$, a has an inverse a^{-1} modulo p.

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Suppose $f(x) = f(y)$ for $x, y \in S$. Then

$$\begin{aligned} ax \equiv ay \pmod{p} &\implies a^{-1}ax \equiv a^{-1}ay \pmod{p} \\ &\implies x \equiv y \pmod{p} \implies x = y. \end{aligned}$$

Thus f is injective (i.e., the same output implies the same input).

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Lemma (exercise; see the note): An injective function from a finite set to itself is bijective.

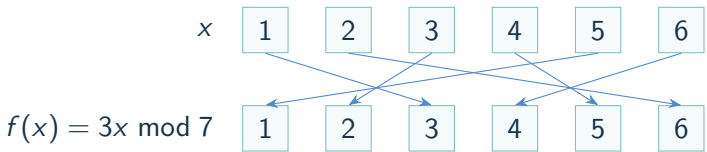
Thus f is bijective.

Fermat's Little Theorem: proof

We have proved that f is a bijection.

Example: $p = 7$, $a = 3$.

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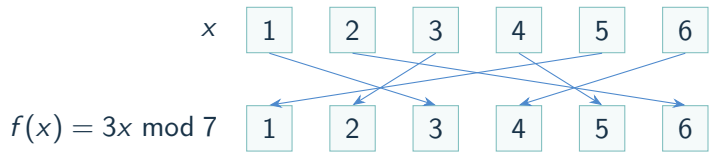


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Now we know:

$$1 \times 2 \times \cdots \times (p-1) = f(1)f(2)\cdots f(p-1).$$

The same product of elements in S, written in a different order.

Fermat's Little Theorem: proof

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Substitute $f(x) \equiv ax \pmod{p}$ into the product identity:

$$(p-1)! \equiv (a \times 1)(a \times 2)\cdots(a \times (p-1)) \pmod{p}.$$

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There are $p-1$ factors of a, so

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Every factor $1, \ldots, p-1$ is coprime to p, so $(p-1)!$ is invertible.

Cancel $(p-1)!$:

$$\boxed{a^{p-1} \equiv 1 \pmod{p}.}$$

Corollary: An inverse from Fermat's little theorem

Corollary

If p is prime and $1 \leq a \leq p-1$, then

$$a^{p-2} \equiv a^{-1} \pmod{p}.$$

That is, a^{p-2} is an inverse of a modulo p.

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If p is prime and $1 \leq a \leq p-1$, then

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That is, a^{p-2} is an inverse of a modulo p.

By Fermat's little theorem,

$$a \times a^{p-2} = a^{p-1} \equiv 1 \pmod{p}.$$

Euler's totient

Definition: Euler's totient

For $m > 1$, $\varphi(m)$ counts how many $x \in \{1, \ldots, m-1\}$ satisfy

$$\gcd(x, m) = 1 \qquad \text{(i.e., } x \text{ is coprime to } m\text{)}.$$

Equivalently, it counts how many of these x have an inverse x^{-1} modulo m.

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For any prime p, $\varphi(p) = p - 1$, since $1, 2, \ldots, p-1$ are all coprime to p.

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Example: $\varphi(15) = 8$.

Euler's theorem

Theorem (Euler)

If $m > 1$ *and* $\gcd(a, m) = 1$, *then*

$$a^{\varphi(m)} \equiv 1 \pmod{m}.$$

The proof is a simple extension of the proof of Fermat's little theorem: use the set of remainders coprime to m.

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The proof is a simple extension of the proof of Fermat's little theorem: use the set of remainders coprime to m.

For any integer $k \geq 0$,

$$\boxed{a^k \equiv a^{k \bmod \varphi(m)} \pmod{m}.}$$

Computing Euler's totient: Two facts

1. Prime powers

For any prime p and integer $k \geq 1$,

$$\varphi(p^k) = p^k - p^{k-1} = (p-1)p^{k-1}.$$

Explanation: Among $1, \ldots, p^k - 1$, all numbers except the multiples $p, 2p, 3p, \ldots$ are coprime to p^k.

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2. Coprime factors

For positive integers a, b with $\gcd(a, b) = 1$,

$$\varphi(ab) = \varphi(a)\varphi(b).$$

Explanation: By CRT, each pair of coprime remainders modulo a and b gives exactly one coprime remainder modulo ab.

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Explanation: By CRT, each pair of coprime remainders modulo a and b gives exactly one coprime remainder modulo ab.

We use $\varphi(1) = 1$. Proofs are in the note.

Computing Euler's totient: General formula

1. Write the prime factorization. For $x > 1$,

$$x = p_1^{e_1} p_2^{e_2} \cdots p_t^{e_t},$$

where the p_i are distinct primes and each $e_i \geq 1$.

Computing Euler's totient: General formula

1. Write the prime factorization. For $x > 1$,

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where the p_i are distinct primes and each $e_i \geq 1$.

2. Apply the two facts. The prime-power factors are pairwise coprime.

Euler's totient formula

$$\begin{aligned}\varphi(x) &= \prod_{i=1}^{t} \varphi(p_i^{e_i}) = \prod_{i=1}^{t} (p_i - 1) p_i^{e_i - 1} \\ &= x \prod_{i=1}^{t} \left(1 - \frac{1}{p_i}\right).\end{aligned}$$

Summary

1. **Modular inverses**
2. **Bézout's identity and extended GCD**
3. **Combining congruences**
   Chinese remainder theorem.
4. **Modular powers**
   Fermat's little theorem and Euler's totient function.

Contents

1. Modular inverses and equations
2. Combining congruences
3. Modular powers: Fermat and Euler
4. Optional: Non-coprime equations
5. Review exercises

When there is no inverse

$\gcd(6, 14) = 2$, so 6 has no inverse modulo 14.

Can either equation still have solutions?

$$6x \equiv 8 \pmod{14} \qquad 6x \equiv 7 \pmod{14}$$

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The second would require $6x - 14k = 7$ for some integer k.

$$\underbrace{6x - 14k}_{\text{even}} \neq \underbrace{7}_{\text{odd}}.$$

No solution to $6x \equiv 7 \pmod{14}$.

Non-coprime case: Reduce the modulus

To solve $ax \equiv b \pmod{m}$, where $m > 1$:

1. Compute $g = \gcd(a, m)$.

Non-coprime case: Reduce the modulus

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2. If $g \nmid b$, there is no solution.
3. If $g \mid b$, divide the coefficient, right-hand side, and modulus by g:

$$\boxed{\frac{a}{g}x \equiv \frac{b}{g} \pmod{m/g}.}$$

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4. Set $m' = m/g$. If $m' > 1$, find

$$x_0 \equiv (a/g)^{-1}(b/g) \pmod{m'}, \qquad 0 \le x_0 < m'.$$

   If $m' = 1$, take $x_0 = 0$.

Non-coprime case: The full solution

Let $g = \gcd(a, m)$ with $g \mid b$, and set $m' = m/g$.

Suppose the reduced congruence gives $x \equiv x_0 \pmod{m'}$.

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All integer solutions

$$x = x_0 + km', \qquad k \in \mathbb{Z}.$$

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The g distinct solutions modulo the original modulus m are

$$\boxed{x \equiv x_0,\ x_0 + m',\ \ldots,\ x_0 + (g-1)m' \pmod{m}.}$$

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If $m' = 1$, take $x_0 = 0$: every integer x is a solution.

Non-coprime case: Apply the reduction

Solve $6x \equiv 8 \pmod{14}$.

$g = \gcd(6, 14) = 2$, and $2 \mid 8$.

Non-coprime case: Apply the reduction

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$g = \gcd(6, 14) = 2$, and $2 \mid 8$. Divide 6, 8, and 14 by 2:

$$6x \equiv 8 \pmod{14} \iff 3x \equiv 4 \pmod{7}.$$

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Since $3^{-1} \equiv 5 \pmod{7}$,

$$x \equiv 5 \times 4 \equiv 6 \pmod{7}.$$

0 1 2 3 4 5 (6) 7 8 9 10 11 12 (13)

$$\boxed{x \equiv 6, 13 \pmod{14}.}$$

Linear congruences: The general rule

Theorem

Let $g = \gcd(a, m)$. *For* $ax \equiv b \pmod{m}$*:*

- *solutions exist exactly when* $g \mid b$*;*
- *if solvable, there are exactly* g *solutions among* $0, \ldots, m-1$.

Linear congruences: The general rule

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Equation	GCD	Number of solutions
$17x \equiv 5 \pmod{43}$	1	1
$6x \equiv 8 \pmod{14}$	2	2
$6x \equiv 7 \pmod{14}$	2	0

Practice: Zero or several solutions?

Solve each congruence:

$$15x \equiv 9 \pmod{24}, \qquad 15x \equiv 10 \pmod{24}.$$

Practice: Zero or several solutions?

Solve each congruence:

$$15x \equiv 9 \pmod{24}, \qquad 15x \equiv 10 \pmod{24}.$$

1. Predict the number of solutions using the GCD.
2. For each solvable equation, list every solution among $0, \ldots, 23$.

Full solutions in the note.

Review exercises: Inverses and congruences

1. Modular inverse. Use extended Euclid to find $12^{-1} \pmod{35}$. Then solve

$$12x \equiv 5 \pmod{35}.$$

Review exercises: Inverses and congruences

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$$12x \equiv 5 \pmod{35}.$$

2. Chinese remainder theorem. Find all integers x satisfying

$$\begin{aligned} x &\equiv 2 \pmod{3}, \\ x &\equiv 3 \pmod{5}, \\ x &\equiv 2 \pmod{7}. \end{aligned}$$

Review exercises: Modular powers

3. Fermat's little theorem. Compute $3^{100} \bmod 7$.

Review exercises: Modular powers

3. **Fermat's little theorem.** Compute $3^{100} \bmod 7$.
4. **Euler's theorem.** Compute $2^{100} \bmod 15$.

Can Euler's theorem also be applied to $3^{100} \bmod 15$? Why?

Review exercises: Modular powers

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5. Challenge. Compute

$$2^{100} \bmod 35.$$

Try both methods:

- Fermat's little theorem and CRT.
- Euler's theorem.